

Multiple integrals – 2

Change of variables and the Jacobian

13 сентября 2026 года

Theoretical minimum.

Theorem 1 (change of variables). If φ is a diffeomorphism of open sets in $\mathbb{R}^n$, then $\int_{\varphi(U)} f = \int_U (f \circ \varphi) |\det D\varphi|$. For a linear map the factor is $|\det A|$, which is exactly the statement that the determinant is the coefficient of volume distortion (Linear algebra – 1).

Fact 1 (the three standard substitutions). Polar: $dx dy = r dr d\varphi$; cylindrical: $dx dy dz = r dr d\varphi dz$; spherical in $\mathbb{R}^3$: $dx dy dz = r^2 \sin \theta dr d\theta d\varphi$, and in $\mathbb{R}^n$ $dx = r^{n-1} dr d\sigma(\omega)$, where σ is the measure on the unit sphere with $\sigma(S^{n-1}) = \frac{2\pi^{n/2}}{\Gamma(n/2)}$. Radial integrals reduce to one dimension at once.

Fact 2 (what polar coordinates give for free). $(\int_{\mathbb{R}} e^{-x^2} dx)^2 = \iint_{\mathbb{R}^2} e^{-x^2-y^2} = \pi$; the volume of the unit ball in $\mathbb{R}^n$ is $\frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}$; and $\int_{\mathbb{R}^n} e^{-\langle Ax, x \rangle} dx = \frac{\pi^{n/2}}{\sqrt{\det A}}$ for positive definite A , by diagonalising A and using the linear change of variables.

Fact 3 (Dirichlet and Liouville). $\int_{x_i \geq 0, \sum x_i \leq 1} x_1^{a_1-1} \dots x_n^{a_n-1} dx = \frac{\Gamma(a_1) \dots \Gamma(a_n)}{\Gamma(a_1 + \dots + a_n + 1)}$, and Liouville's formula extends this to $\int f(x_1 + \dots + x_n) \prod x_i^{a_i-1} dx$. Every “integral over a simplex or an octahedron” problem is one of these two.

1. Prove the change of variables formula for a linear map from the definition of the determinant as a volume coefficient, deduce the polar and spherical formulas, and use them to compute $\int_{\mathbb{R}} e^{-x^2} dx$, the volume of the unit ball in $\mathbb{R}^n$ and the area of S^{n-1} .
2. Prove Fact 3 and use it to compute $\int_{\|x\|_1 \leq 1} |x_1 x_2 \dots x_n| dx$ and the volume of the set $\{x_i \geq 0, x_1^p + \dots + x_n^p \leq 1\}$ for $p > 0$.

3. (Putnam 2021) Let

$$I(R) = \iint_{x^2+y^2 \leq R^2} \left(\frac{1+2x^2}{1+x^4+6x^2y^2+y^4} - \frac{1+y^2}{2+x^4+y^4} \right) dx dy.$$

Find $\lim_{R \rightarrow \infty} I(R)$ or show that this limit does not exist.

4. (Kapačëв 1.96) Compute $\iint_{x^2+y^2 \leq 1} e^x \cos y \, dx \, dy$, and explain the answer through the mean value property of harmonic functions (Complex analysis – 8).
5. (VJIMC 2018) Compute the integral $\iint_{\mathbb{R}^2} \left(\frac{1 - e^{-xy}}{xy} \right)^2 e^{-x^2-y^2} \, dx \, dy$.